

- Q.28 Describe structure of multimode step index fiber?
- Q.29 List any four comparisons between PIN diode & APD.
- Q.30 Differentiate between step index and graded index fiber.
- Q.31 Explain the working of directional coupler.
- Q.32 What is photomultiplier? Explain.
- Q.33 Write a note on attenuators.
- Q.34 Write applications of optical fiber.
- Q.35 Write a note on distributed feedback laser diode.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is the principle of laser? What are different types of lasers?
- Q.37 Explain the Electromagnetic spectrum used in optical fibre communication.
- Q.38 Explain numerical aperture with the help of suitable diagram. What is the importance of numerical aperture in optical fiber communication?

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5th Sem / Eltx. / Power Eltx. Sub : Optical Fiber Communication

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The bandwidth of a fiber is limited by ____
a) Mode b) Wavelength
c) Dispersion d) Frequency
- Q.2 What are two types of optical detectors
a) LED's & APDs
b) PIN diode & APDs
c) APDs & Laser diodes
d) Laser diodes & PIN diodes
- Q.3 The spreading of a light pulse as it travels along the fiber is caused by what mechanism ____.
a) Diffraction b) Attenuation
c) Dispersion d) Absorption
- Q.4 Fiber Optics used which medium to send information ____.
a) Light b) Water
c) Electrons d) Phonons

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Q.5 Light travelling in optical fiber follows which of the following principles.

- a) Huygen's principle b) Reflection theory
- c) Light theory d) All

Q.6 In optical fibers, light propagates mainly along the _____.

- a) Cladding b) Core
- c) Buffer d) Jacket

Q.7 The attenuation losses are measured in terms of _____

- a) Bel b) Decibel
- c) Del d) None

Q.8 The refractive index of the core in comparison of cladding is _____.

- a) Higher b) Lower
- c) Same d) None

Q.9 The light emitted diode is usually made from _____

- a) Gallium arsenide
- b) Silicon
- c) Aluminium
- d) Germanium

Q.10 In spontaneous emission _____ are emitted in random manner

- a) Electrons b) Photons
- c) Protons d) None

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 PVC stands for _____?

Q.12 Unit of dispersion is _____?

Q.13 SOA stands for _____?

Q.14 LED stands for _____.

Q.15 APD stands for _____.

Q.16 LED is a coherent source of light (True / False)

Q.17 _____ responsible for blue colour of the sky.

Q.18 Define Bit rate.

Q.19 Unit of frequency is _____.

Q.20 WDM _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Draw and explain basic structure of optical fiber.

Q.22 Define Snells law.

Q.23 What is principle of photo detection.

Q.24 Write short note on bending losses.

Q.25 Explain the operating characteristics of LED.

Q.26 Write short note on Avalanche Photo diode (APD).

Q.27 What are the different types of connectors used in optical fibre communication?

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